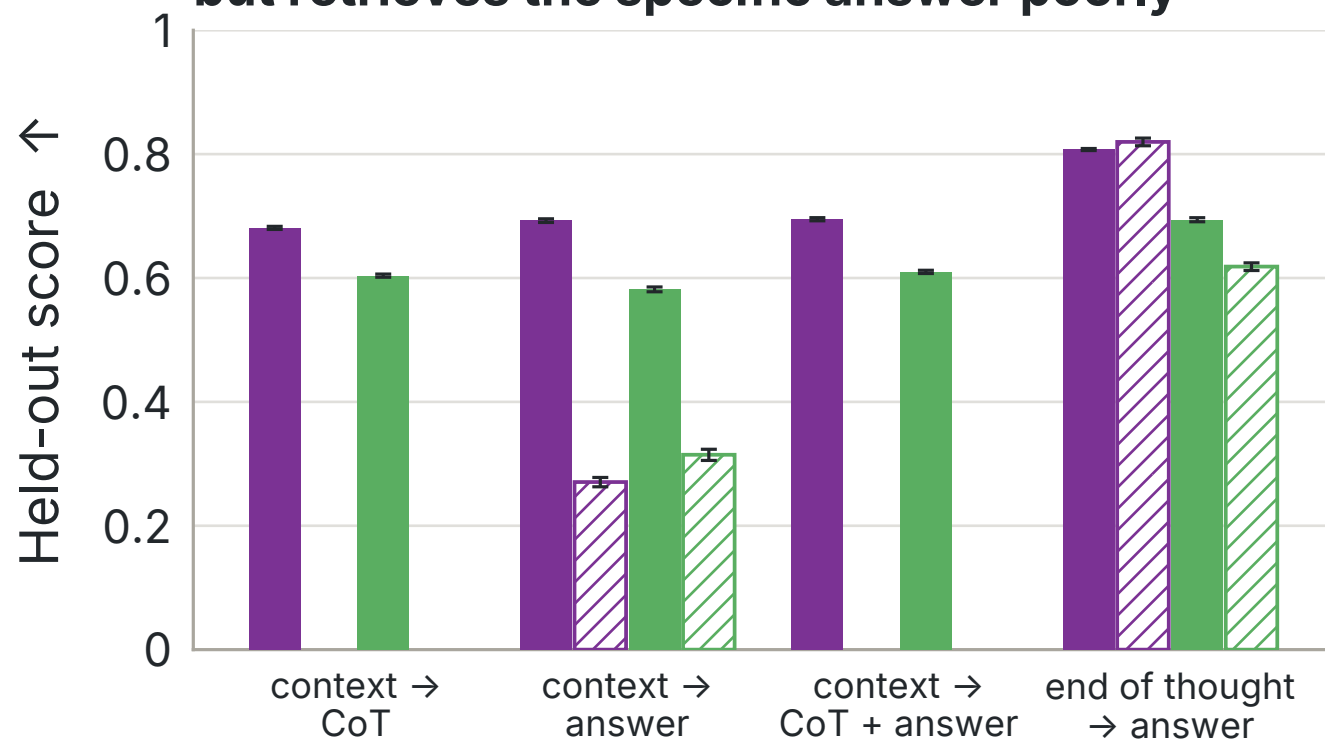


CORPORA

Needs-reasoning corpora No-reasoning corpora

A · OPENTHINKER3-7B, LAYER 19

The context state predicts the answer state, but retrieves the specific answer poorly

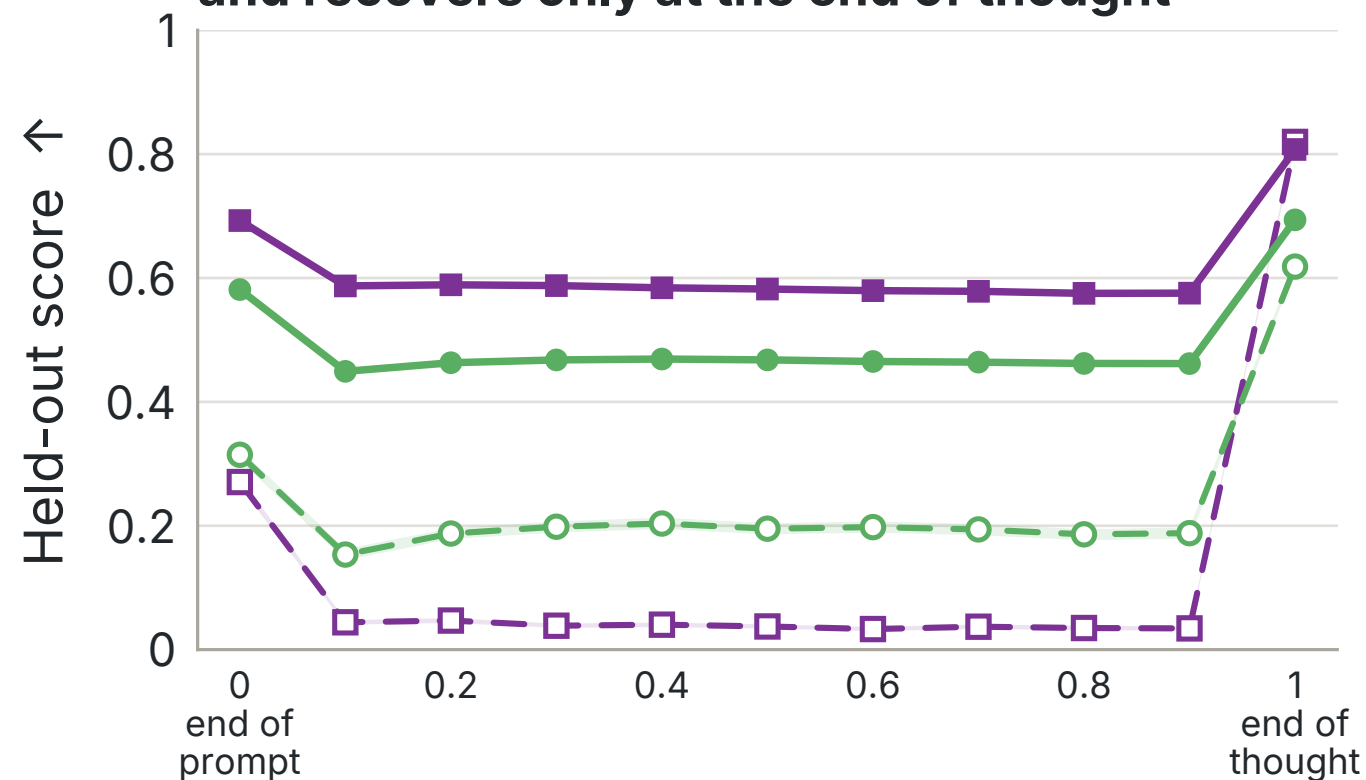


METRIC

Held-out R^2 Answer retrieval (lift over chance)

B · OPENTHINKER3-7B, LAYER 19

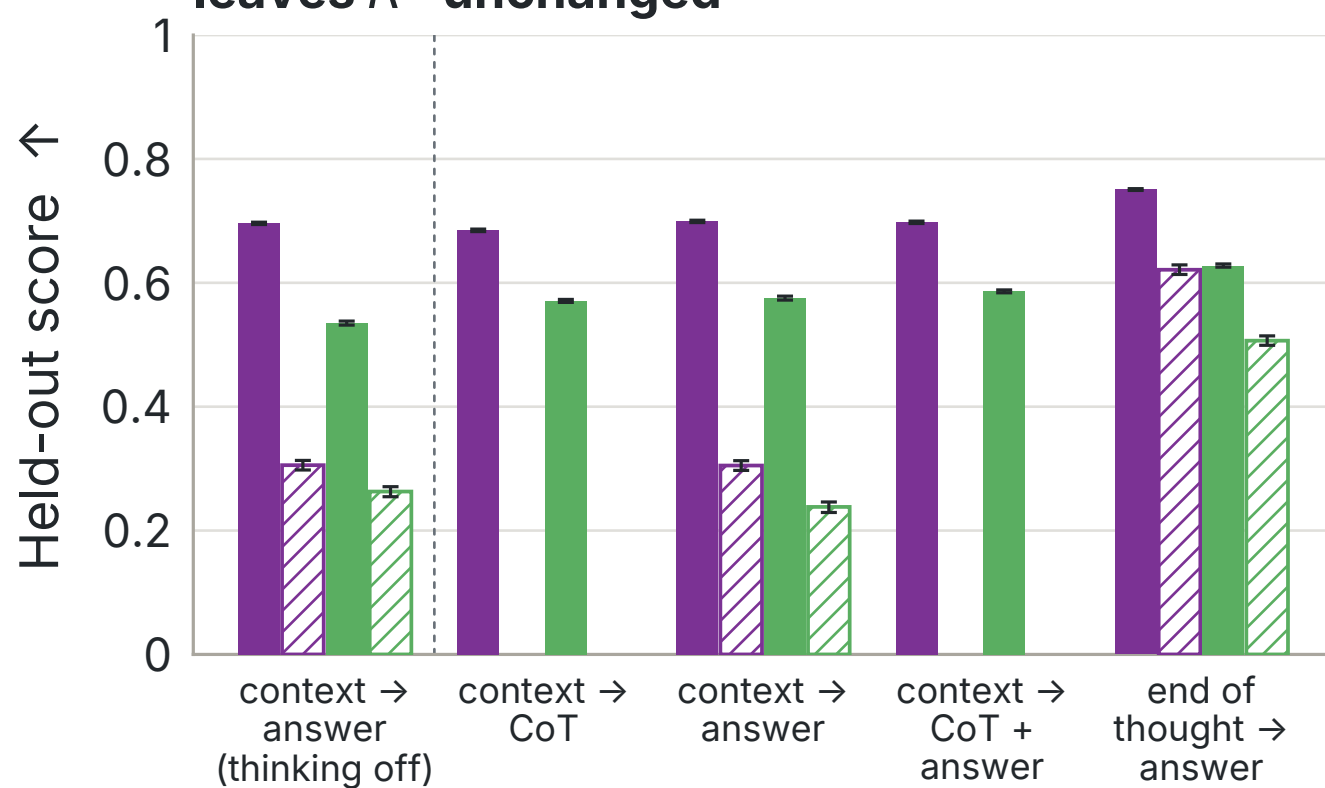
Predictability dips inside the reasoning trace and recovers only at the end of thought



Position in the thinking span

C · QWEN3-8B, LAYER 24

Disabling thinking on the same weights leaves R^2 unchanged



D · QWEN2.5-7B-INSTRUCT (BEFORE) → OPENTHINKER3-7B (AFTER), LAYER 19

The context state before reasoning training predicts the answer state after it

